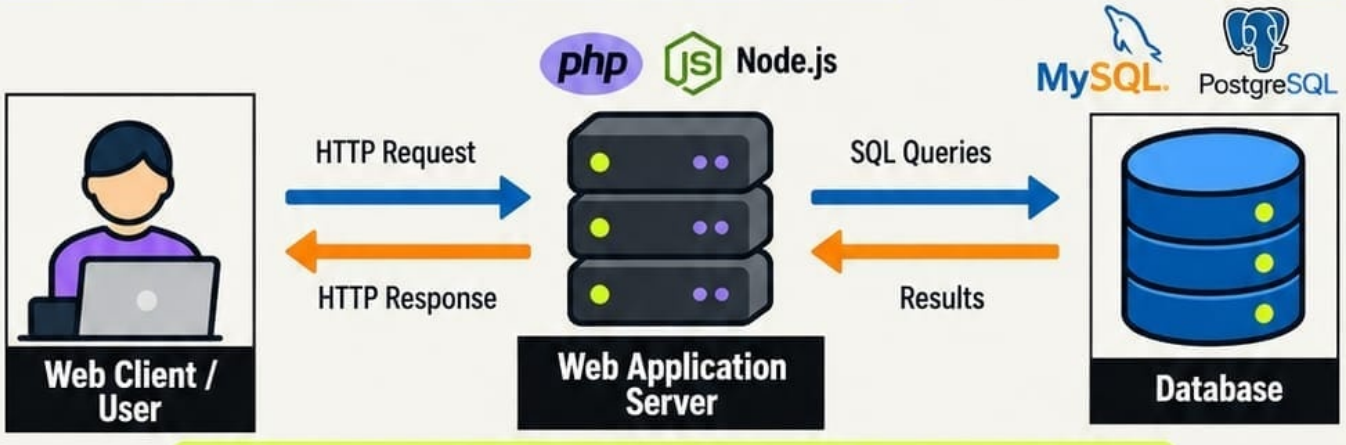
Name: MEET RAMANI | Roll No.: 25DCE100 | Date: _____

WEB DATABASES: Storing Application Data

Web Development Frameworks (ITUE203) | Unit 7: Connect Web Application Database

DATABASE CONNECTIVITY



The server connects to the database using a driver or database library.

DATABASE TYPES

Relational
Structured tables; SQL

NoSQL
Flexible data models

Key-Value
Key maps to a value

Document
JSON-like documents

RELATIONAL TABLES

Users (Parent Table)

ID	Name	Email
1	Asha	asha@example.com
2	Ravi	ravi@example.com

Columns (fields) / Rows (records)

Relationship via user_id

Orders (Child Table)

ID	User_ID	Product	Amount
101	1	Book	500
102	1	Pen	50
103	2	Laptop	75000

Columns (fields) / Rows (records)

COMMON SQL QUERIES

CREATE TABLE — Define structure

```
CREATE TABLE users (
  id INT PRIMARY KEY,
  name VARCHAR(100),
  email VARCHAR(100)
);
```

INSERT — Add records

```
INSERT INTO users
(name, email)
VALUES ('Asha',
 'asha@example.com');
```

SELECT + WHERE — Retrieve / filter records

```
SELECT * FROM users
WHERE id = 1;
```

UPDATE — Modify records

```
UPDATE users
SET email = 'ravi@ex.com'
WHERE id = 2;
```

DELETE — Remove records

```
DELETE FROM users
WHERE id = 1;
```

JOIN — Combine related tables

```
SELECT u.name, o.product
FROM users u
JOIN orders o
ON u.id = o.user_id;
```

WHY CONNECTIVITY MATTERS

Connects user interfaces to persistent, structured data for dynamic web applications.